A triangle that does not contain a right angle is called an **oblique triangle**.

An oblique triangle will have either three acute angles or one obtuse angle and two acute angles.



Acute Triangle
All angles acute



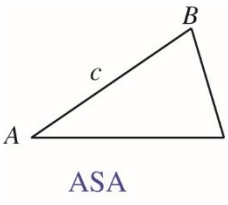
Obtuse Triangle
One angle obtuse

For an oblique triangle, given the length of a side and any other two parts of the triangle we can solve the triangle.

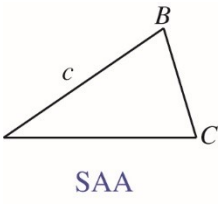
There are four distinct cases involving known angles (A) and known sides (S).

Solved with Law of Sines

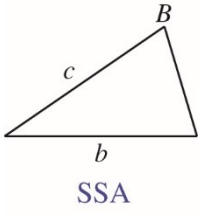
Two angles and any side (ASA or SAA)
Two sides and an angle opposite one of them (SSA)



ASA



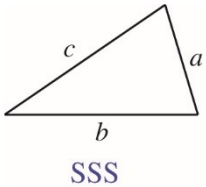
SAA



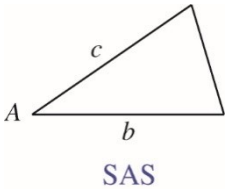
SSA

Solved with Law of Cosines

Three sides (SSS)
Two sides and the angle between them (SAS)



SSS



SAS

Solve a Triangle Using the Law of Sines (SAA or ASA)

Law of Sines If ΔABC is a triangle with sides of length a, b, and c opposite angles A, B, and C, respectively, then

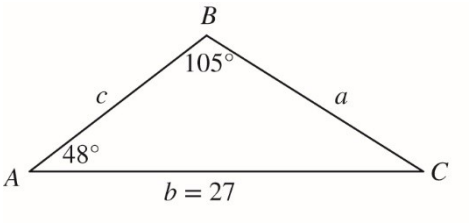
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}.$$

The ratio of the length of a side of a triangle to the sine of the angle opposite that side is the same for all three side-angle combinations.

To apply the law of sines, we need to have one complete ratio and one other piece of information about the triangle (another side or another angle).

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

1. Solve the triangle. Round the lengths of sides and measures of the angles to 1 decimal place if necessary.



$$C = 180° - 105° - 48° = 27°$$

Step 1: Find the exact measure of the missing angle.

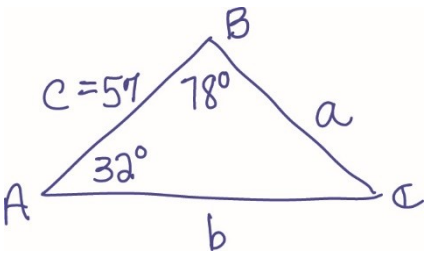
Step 2: Write one complete ratio and use the third piece of information to fill in part of a ratio.

$$\frac{a}{\sin 48°} = \frac{27}{\sin 105°} \rightarrow a = \frac{27 \sin 48°}{\sin 105°} \approx 20.8$$

Step 3: Choose another set of ratios to solve, preferably one with no rounded values.

$$\frac{c}{\sin 27°} = \frac{27}{\sin 105°} \rightarrow c = \frac{27 \sin 27°}{\sin 105°} \approx 12.7$$

2. Solve the triangle with $A = 32°$, $B = 78°$, and $c = 57$. Round the lengths of the sides to 1 decimal place.



① $C = 180° - 78° - 32° = 70°$

② $$\frac{b}{\sin 78°} = \frac{57}{\sin 70°} \rightarrow b = \frac{57 \sin 78°}{\sin 70°} \approx 59.3$$

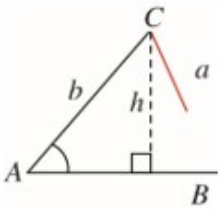
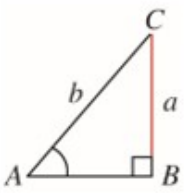
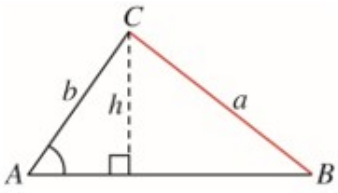
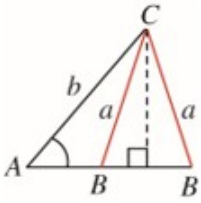
③ $$\frac{a}{\sin 32°} = \frac{57}{\sin 70°} \rightarrow a = \frac{57 \sin 32°}{\sin 70°} \approx 32.1$$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

Solve a Triangle Using the Law of Sines (SSA) Ambiguous Case

Suppose that the measure of angle A and the lengths of sides a and b of a triangle are given.

Depending on the length of side a (shown in red) relative to the length of the altitude h, we have four different scenarios.

No triangle	One right triangle	One oblique triangle	Two triangles: One acute and one obtuse triangle
			
$a < h$ Side a is too short to meet side c.	$a = h$ Side a exactly equals the altitude	$a > b$ One triangle possible	$h < a < b$ Side a intersects side c at an obtuse or acute angle.

---

3. (No solution) Solve the triangle with $A = 28°$, $a = 12$, and $b = 38$. Round the measures of the unknown angles and side to 1 decimal place.

$$\frac{\sin B}{38} = \frac{\sin 28°}{12} \rightarrow \sin B = \frac{38 \sin 28°}{12} \approx 1.5 > 1 \text{ no triangle}$$

4. (1 solution) Solve the triangle with $C = 68°$, $b = 13$, and $c = 15$. Round the measures of the unknown angles and side to 1 decimal place.

$$\frac{\sin B}{13} = \frac{\sin 68°}{15} \rightarrow \sin B = \frac{13 \sin 68°}{15}$$

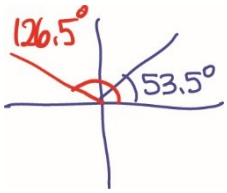
$$B = \sin^{-1}\left(\frac{13 \sin 68°}{15}\right) \approx 53.5° \text{ or } 126.5°$$

if $B = 53.5°$

$$A = 180° - 68° - 53.5° = 58.5°$$

$$\frac{a}{\sin 58.5°} = \frac{15}{\sin 68°}$$

$$a = \frac{15 \sin 58.5°}{\sin 68°} \approx 13.8$$



if $B = 126.5°$

$$A = 180° - 68° - 126.5° = -14.5°$$

No

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

5. (2 solutions) Solve the triangle with $B = 28°$, $b = 10$, and $a = 18$. Round the measures of the unknown angles and side to 1 decimal place.

$$\frac{\sin A}{18} = \frac{\sin 28°}{10} \rightarrow \sin A = \frac{18 \sin 28°}{10} \rightarrow A = \sin^{-1}\left(\frac{18 \sin 28°}{10}\right)$$
$$\approx 57.7° \text{ or } 180° - 57.7°$$
$$= 122.3°$$

if $A = 57.7°$, $C = 180° - 28° - 57.7°$
$= 94.3°$

$$\frac{c}{\sin 94.3°} = \frac{10}{\sin 28°}$$
$$c = \frac{10 \sin 94.3°}{\sin 28°} \approx 21.2$$

if $A = 122.3°$, $C = 180° - 28° - 122.3°$
$= 29.7°$

$$\frac{c}{\sin 29.7°} = \frac{10}{\sin 28°}$$
$$c = \frac{10 \sin 29.7°}{\sin 28°} \approx 10.6$$

$a = 18,\quad b = 10,\quad c = 21.2,\quad A = 57.7°,\quad B = 28°,\quad C = 94.3°$

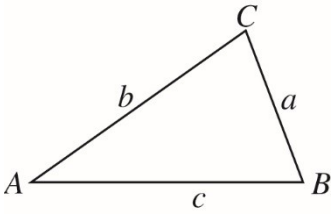
Triangle # 1:

Triangle # 2: $a = 18,\quad b = 10,\quad c = 10.6,\quad A = 122.3°,\quad B = 28°,\quad C = 29.7°$

Compute the Area of a Triangle given SAS

For a triangle ΔABC, the area of the triangle is given by

$$\text{Area} = \frac{1}{2}bc \sin A \quad \text{or} \quad \text{Area} = \frac{1}{2}ab \sin C \quad \text{or} \quad \text{Area} = \frac{1}{2}ac \sin B.$$



---

6. Find the area of a triangle with the given measurements. Round to one decimal place.

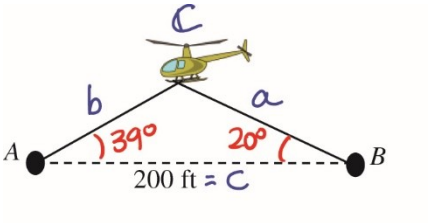
$A = 87°$, $b = 10$ ft, $c = 23$ ft

$$\text{Area} = \frac{1}{2}(10)(23) \sin 87° \approx 114.8 \text{ ft}^2$$

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.

Apply the Law of Sines

7. A helicopter is on a path directly overhead line AB when it is simultaneously observed from locations A and B separated by 200 ft. If the angle of elevation from A is 39° and the angle of elevation from B is 20°, what is the distance from each location to the helicopter? Round to the nearest foot.



$A = 39°, B = 20°$

$C = 180° - 39° - 20° = 121°$

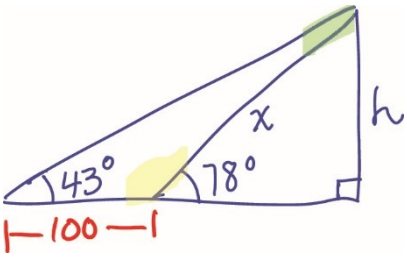
side $c = 200$

$$\frac{b}{\sin 20°} = \frac{200}{\sin 121°} \rightarrow b = \frac{200 \sin 20°}{\sin 121°} \approx 80$$

$$\frac{a}{\sin 39°} = \frac{200}{\sin 121°} \rightarrow a = \frac{200 \sin 39°}{\sin 121°} \approx 147$$

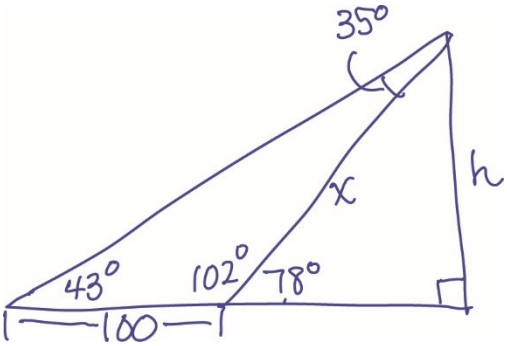
dist. from A to heli. = 80 ft.
dist. from B to heli. = 147 ft.

8. To estimate the height of a building, two measurements are taken. The first measurement shows an angle of elevation to the top of the building as 43°. The second measurement, taken 100 feet closer to the base, yields an angle of elevation of 78°. From these measurements, estimate the height of the building. Round to the nearest foot.



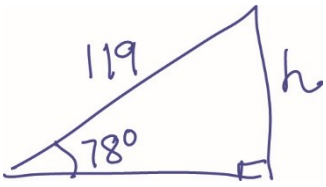
$180° - 78° = 102°$

$180° - 43° - 102° = 35°$



find x first

$$\frac{x}{\sin 43°} = \frac{100}{\sin 35°} \rightarrow x = \frac{100 \sin 43°}{\sin 35°} \approx 119$$



Building: 116 ft

© McGraw Hill LLC. All rights reserved. No reproduction or distribution without the prior written consent of McGraw Hill LLC.